INCH-POUND
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DETAIL SPECIFICATION

LEATHER, CATTLEHIDE, RIGGING

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for rigging leather made from full grain cattlehide.

1.2 Classification. The leather is available in the following types and classes:

1.2.1 Types.

Type I - Sides
Type II - Backs

1.2.2 Classes.

Class 1 – Mildew resistant treated
Class 2 – Untreated

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AMSC N/A FSC 8330

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. N/A

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Federal Insecticide, Fungicide and Rodenticide (FIFRA)
(40 CFR part 162 State Registration of Pesticides Products)

Federal Insecticide, Fungicide and Rodenticide (FIFRA)
as amended by the Food Quality Protection Act of 1996 and the Pesticide Registration Improvement Act of 2003

(Copies of these documents are available online at https://www.epa.gov/pesticides.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles

(Copies of this document are available on line at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

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ASTM INTERNATIONAL

ASTM D1813 - Standard Test Method for Measuring Thickness of Leather Specimens
ASTM D1814 - Standard Test Method for Measuring Thickness of Leather Units
ASTM D2209 - Standard Test Method for Tensile Strength of Leather.
ASTM D2617 - Standard Test Method for Total Ash in Leather
ASTM D2810 - Standard Test Method for pH of Leather
ASTM D2813 - Standard Practice for Sampling Leather for Physical and Chemical Tests
ASTM D3495 - Standard Test Method for Hexane Extraction of Leather
ASTM D6075 - Standard Test Method for Shrinkage Temperature of Leather

(Copies of these documents are available online at https://www.astm.org.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.routledge.com.)

INTERNATIONAL ORGANIZATION for STANDARDIZATION (ISO)

ISO 13365-1 - Leather – Chemical Determination of the Preservative (TCMTB, PCMC, OPP, OIT) Content in Leather by Liquid Chromatography Part 1 – Acetonitrile Extraction Method

(Copies of this document are available online at https://www.iso.org.

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials. The strap leather shall be full grain cattlehide. Bull hides will not be acceptable. The flesh side of the leather shall be smooth and free of coarse loose fibers.

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3.4 Tannage. The leather shall be vegetable tanned.

3.5 Color. Unless otherwise specified (see 6.2), the color of the leather shall be natural as produced by the tanning agents. If necessary, a light application of dye or pigment may be added to level the color of the grain side (see 6.3). Visual examination of the color and appearance shall be as specified in 4.6.1.

3.6 Physical requirements.

3.6.1 Trim (form and area). The leather shall be furnished in the form of sides or backs as required for the type specified (see 6.2). The area expressed in terms of square feet shall be legibly marked in the butt area on the flesh side of the leather.

3.6.2 Cracking. The leather shall not crack when tested as specified in 4.5.

3.6.3 Weight and thickness. The leather shall be of the weight and thickness specified (6.2) as shown on Table I when tested as specified in 4.5. The leather may be shaved on the flesh to provide greater uniformity of thickness but shall not be split (see 6.4).

TABLE I. Weight and thickness requirements.

Weight	Thickness (ounces) 1/
Light	6 - 7.75
Medium	8 - 9.75
Heavy	10 or greater

1/ Note: 1 ounce = 1/64 inch)

3.6.4 Breaking force. Not less than 80 percent of the specimens tested shall meet the minimum breaking force values in Table II when tested as specified in 4.5. Not less than 80 of the specimens tested shall have elongation values of 25 percent of less at the loads shown in Table II when tested as specified in 4.5.

TABLE II. Breaking force and elongation requirements.

Thickness	Breaking force, pounds (min.) (ounces)	Elongation load (pounds)
6 - 7.75	150	100
8 - 9.75	200	150
10 or greater	250	200

3.7 Chemical requirements. The leather shall conform to the chemical requirements of Table III when tested as specified in 4.5.

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TABLE III. Chemical requirements.

Characteristic	Requirement
Hexane soluble material, percent (max.) 1/ Minimum Maximum	 7.0 18.0
Water soluble material, percent (max.) 1/	18.0
Total ash, percent (max.) 1/	5.0
Acidity (pH)	3.3 – 5.0

1/ Calculated on a moisture-free basis.

3.7.1 Fungicide (mildew resistance), Class 1 only. Unless otherwise specified (see 6.2), the leather shall be treated with one of the following fungicides to provide mildew resistance. The fungicides to treat the leather shall be an EPA registered product that complies with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) (see 2.2.2).

3.7.1.1 2-(Thiocyanomethylthio) bensothiazole (TCMTB). The leather shall contain not less than 100 ppm and not more than 400 ppm of active ingredient TCMTB when tested as specified in 4.5.

3.7.1.2 *Ortho*-phenyl phenol (OPP). The leather shall contain not less than 540 ppm and not more than 2580 ppm of active ingredient OPP when tested as specified in 4.5

3.7.1.3 Diiodomethyl-*para*-tolylsulfone (DIMPTS). The leather shall contain not less than 300 ppm and not more than 2000 ppm of active ingredient DIMPTS when tested as specified in 4.5.

3.9 Workmanship. The finished leather shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.3 and 4.4 shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5 and 4.7.

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4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination of the end item. The leather for each type shall be examined and defects shall be classified as stated in Table IV. The sample unit for these examinations shall be one (1) side or back. The lot size shall be expressed in terms of one (1) side or back. The contract or purchase order shall identify the number of samples required for thickness for acceptance and rejection of the leather skins. Separate examinations shall be performed for visual and the thickness, but the same sample may be used for each examination.

TABLE IV. Visual examination defects.

Examination	Defect 1/
Material	Not full grain cattlehide
Trim	Not sides or backs as required
Quality of leather	Burn mark on flesh
	Flesh side not smooth or has areas of loose flesh in excess of four (4) square inches in total area.
	Any hole, deep cut, scratch, damaged hipmark, rough shoulder, stain, or other defects which individually or collectively impair the cutting value of more than 10 percent of the total area of the back or side. Raw streak..
Color (all types)	Unless otherwise specified, color not natural
Area (expressed in terms of square feet)	Not as marked on leather
Marking	Omitted, not readable
Thickness	Not as specified 2/

1/ Light, well healed scratches and grub holes, light fat wrinkles, and slight stains shall not be classified as defects.
2/ Thickness shall be measured in accordance with ASTM D1814.

4.5 End item testing. The finished leather shall be tested for the characteristics as specified in Table V. The methods of testing as specified in Table V shall be followed. Sample grinding and sampling location from which the sample unit is to be obtained shall be in accordance with ASTM D2813. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any of the following conditions exists:

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a. More than three (3) failures occur for breaking force and elongation,
b. More than one (1) specimen fails to meet any of the remaining requirements applicable to the sample unit.
c. Any composite fails to meet the specified requirement.

TABLE V. End item tests.

Characteristic	Requirement Reference	Test Method	Requirement	
			Sample unit	Composite sample
Material identification	See 3.3	Visual (see 6.4)		-
Tannage	See 3.4	Visual (see 6.4)		-
Color	See 3.5	4.6.1		-
Cracking	3.6.2	ASTM D6075	X	-
Thickness	3.6.3 &Table I	ASTM D1813	X	-
Breaking force	3.6.4 & Table II	ASTM D2209	X	-
Elongation	3.6.4 & Table II	ASTM D2209	X	-
Hexane-soluble material	Table III	ASTM D3495	-	X
Water-soluble material	Table III	ASTM D3495	-	X
Total ash	Table III	ASTM D2617	-	X
Acidity (pH)	Table III	ASTM D2810	-	X
Fungicide (Class 1)	See 3.7.1	ISO 13365-1 2/	-	X

1/ Evaluated using AATCC EP8.
2/ When testing DIMPTS, use an HPLC analytical standard to measure the concentration of DIMPTS in accordance with ISO 13365-1.

4.6 Methods of testing and inspection.

4.6.1 Visual shade examination. The finished leather shall be viewed using AATCC Evaluation Procedure 9, Option C when viewed under filtered tungsten lamps which approximate artificial daylight D_{75} illuminant having a correlated color temperature of 7500K (± 200 K) with illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2850 K (± 200 K).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the

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Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The strap leather is intended primarily for use in shoes and textile repair shops, trailer mounted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type, class and subclass required (see 1.2)
c. The specific issue of individual documents referenced
d. When first article is required (see 3.1)
e. Color required (see 3.5)
f. Conformance inspection (see 4.3)
g. Inspection conditions (see 4.3.1)
h. Packaging (see 5.1)

6.3 Standard sample. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Subject term (key word) listing.

Backs
Fungicide
Mildew resistant
Sides

6.6 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CIVIL AGENCY COORDINATING ACTIVITY
GSA-FAS

Custodians:
Army - GL
Navy - NU

Preparing Activity
DLA-CT

(Project: 8330-2022-003)

Review activities:
Army - MD

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